

5(a)	work done per unit charge	B1
	(work done on charge) moving positive charge from infinity	B1
5(b)(i)	$(2.0 \times 10^{-9}) / 4\pi\epsilon_0(4.0 \times 10^{-2}) + Q / 4\pi\epsilon_0(8.0 \times 10^{-2}) = 0$	C1
	$Q = 4.0 \times 10^{-9} \text{ C}$	A1
	Q given with negative sign	B1
5(b)(ii)	change = 1200 V	A1
5(c)	$\frac{1}{2}mv^2 = qV$	C1
	$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 1200$	C1
	$v = 3.4 \times 10^5 \text{ m s}^{-1}$	A1